VCS Screen

Visual Contrast Sensitivity Screening Report

Concept: Dr Cameron Jones, PhD · www.biologicalhealthservices.com.au

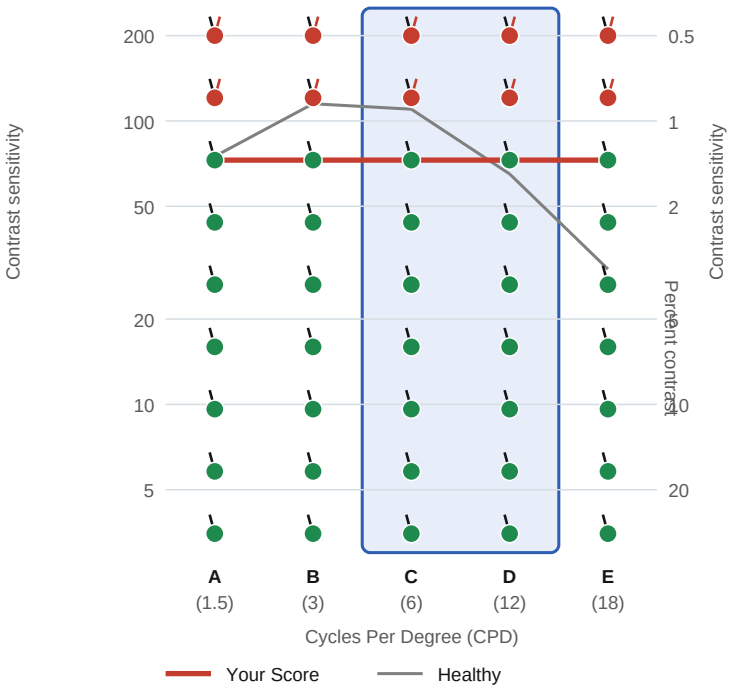
VCS Test Results

Test Identifier: muyhb2f4
Test Type: Sine-wave grating, 5 × 9, full mode
Calibration: 50 cm · 40.0 px/cm · 35.0 px/deg

Test Date: 10/8/2026, 6:06:40 AM
Name: _____

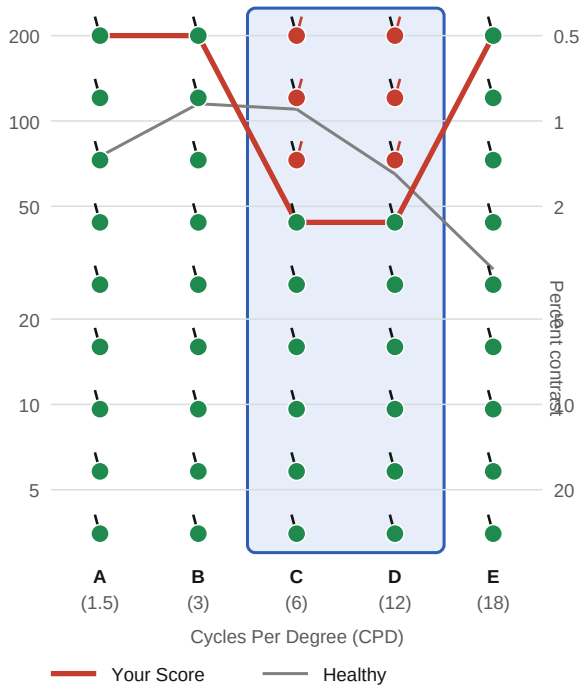
RESULTS: POSITIVE · TOTAL SCORE: 74/90 (82%) · BIOTOXIN SCORE: 26/36 (72%)

Left Eye



Total Score: 35/45 (78%)
Biotoxin Score: 14/18 (78%)

Right Eye



Total Score: 39/45 (87%)
Biotoxin Score: 12/18 (67%)

The charts above plot the results of your contrast sensitivity screen for each eye. The left axes show contrast sensitivity, the right axes percent contrast, and the bottom axes the spatial frequency of the bars in cycles per degree (CPD); the closer the bars, the higher the spatial frequency.

Each circle is one test image, placed by its contrast level and spatial frequency. Columns C and D (6 and 12 CPD) are outlined in blue. A green circle means you identified the tilt correctly; a red circle means you did not. Black tails show the tilt the image actually had; red tails show the tilt you chose.

The grey line is a typical contrast sensitivity curve for healthy adults on FACT-style charts. The red line joins the faintest image you identified correctly in each column. Higher is better; a red line that dips well below the grey line at any point is worth discussing with your healthcare provider.

RESULTS - OVERALL: POSITIVE

You identified the tilt of the bars 74 times out of 90, for a Total Score of 74 (82%). Your right eye performed better than the other. A score below 100% means at least one image was missed; look at where the misses fall before drawing any conclusion.

RESULTS - BIOTOXIN COLUMNS (C and D): FLAGGED

In columns C and D (6 and 12 CPD), the columns most often reported as affected in biotoxin-associated illness, you answered correctly 26 times out of 36 (72%). A deficit here is consistent with, but not specific to, biotoxin exposure; it is also produced by refractive error, cataract, other eye disease, alcohol, tobacco and certain medications. Consider an environmental assessment of your home or workplace and discuss these results with a clinician familiar with contrast sensitivity testing.

RESULTS - COLUMN E (18 CPD): CLEAR

Threshold sensitivity at 18 CPD was 200 (right) and 73 (left), against a typical healthy value of about 30. Contrast function at 18 CPD is often used to track progress during treatment because it tends to recover first.

Threshold sensitivity by column

	A (1.5)	B (3)	C (6)	D (12)	E (18)	Score
Right eye	200	200	44	44	200	39/45
Left eye	73	73	73	73	73	35/45
Healthy (typical)	75	115	110	65	30	

Screening only. This report is generated by a web page and is not a diagnosis. Results depend on the display, its brightness setting, viewing distance and ambient light; the lowest one or two contrast rows exceed what most 8-bit displays can render precisely. Compare repeat tests taken on the same screen and settings. Chart layout and interpretation follow the sources listed under References.

References

1. Ginsburg AP. A new contrast sensitivity vision test chart. Am J Optom Physiol Opt. 1984;61(6):403-407.

2. Ginsburg AP. Contrast sensitivity, drivers' visibility, and vision standards. Transportation Research Record. 1987;1149:32-39.

3. Ginsburg AP. Contrast sensitivity and functional vision. Int Ophthalmol Clin. 2003;43(2):5-15.

4. Hudnell HK, House D, Schmid J, Koltai D, Stopford W, Wilkins J, Savitz DA, Swinker M, Music S. Human visual function in the North Carolina clinical study on possible estuary-associated syndrome. J Toxicol Environ Health A. 2001;62(8):575-594. doi:10.1080/15287390151079633

5. Shoemaker RC, Hudnell HK. Possible estuary-associated syndrome: symptoms, vision, and treatment. Environ Health Perspect. 2001;109(5):539-545.

6. Hudnell HK. Chronic biotoxin-associated illness: multiple-system symptoms, a vision deficit, and effect of cholestyramine. Neurotoxicol Teratol. 2005;27(5):733-743.

7. Shoemaker RC, House DE. A time-series study of sick building syndrome: chronic, biotoxin-associated illness from exposure to water-damaged buildings. Neurotoxicol Teratol. 2005;27(1):29-46.

8. Shoemaker RC, House DE. Sick building syndrome (SBS) and exposure to water-damaged buildings: time series study, clinical trial and mechanisms. Neurotoxicol Teratol. 2006;28(5):573-588.

9. McMahon SW, Kundomal KA, Yangalasetty S. Pediatrics norms for visual contrast sensitivity using an APT VCS tester. Med Res Arch. 2017;5(5).

10. Pelli DG, Bex P. Measuring contrast sensitivity. Vision Res. 2013;90:10-14.

11. Owsley C. Contrast sensitivity. Ophthalmol Clin North Am. 2003;16(2):171-177.

12. Thomas G, Burton NC, Mueller C, Page E, Vesper S. Comparison of work-related symptoms and visual contrast sensitivity between employees at a severely water-damaged school and a school without significant water damage. Am J Ind Med. 2012;55(9):844-854. doi:10.1002/ajim.22059

13. Jimenez-Barbosa IA, Di Lizio S, Ahn SB, et al. Assessment of visual contrast sensitivity in biotoxin-exposed individuals using four testing methods. Ann Med. 2026;58(1):2646821. doi:10.1080/07853890.2026.2646821

14. American College of Medical Toxicology. Position statement: medical toxicology considerations in the diagnosis and treatment of patients with concerns about mold-related inhalation exposures. 2025.

15. Denning DW, Chakrabarti A. Pulmonary and sinus fungal diseases in non-immunocompromised patients. Lancet Infect Dis. 2017;17(11):e357-e366. doi:10.1016/S1473-3099(17)30309-2

16. Valtonen V. Clinical diagnosis of the dampness and mold hypersensitivity syndrome: review of the literature and suggested diagnostic criteria. Front Immunol. 2017;8:951. doi:10.3389/fimmu.2017.00951

Evidence status

VCS measures contrast discrimination across spatial frequencies, reflecting visual processing in the central nervous system rather than the optics of the eye. Shoemaker popularised it as a bedside marker of biotoxin-related illness after exposure to water-damaged buildings. The evidence is mixed and the test is not validated as a diagnostic tool for mould-related illness.

A NIOSH study (ref 12) found lower VCS values at every spatial frequency among employees of a severely water-damaged school than at a comparison school, with more rash, respiratory and constitutional symptoms, yet advised against recommending VCS for mould-exposed people because earlier studies used non-representative samples, lacked comparison groups, characterised exposure poorly and were confounded by conditions such as fibromyalgia and chronic fatigue. A 2026 four-method comparison (ref 13) found the handheld Shoemaker chart did not separate exposed people from controls while digital tests did (best: 100% sensitivity, 60-80% specificity); the digital tools remain investigational.

The American College of Medical Toxicology (ref 14) states there is no validated biomarker for mould or mycotoxin exposure; a Lancet Infectious Diseases review (ref 15) notes no diagnostic test exists for building-related illness, which rests on history, exposure assessment and exclusion of other causes; a proposed framework for dampness and mould hypersensitivity syndrome (ref 16) uses clinical and exposure criteria only. In an Australian clinical context VCS should be treated as non-validated and not a stand-alone diagnostic or monitoring tool. Reduced contrast sensitivity is also produced by refractive error, cataract, glaucoma, macular disease, diabetes, alcohol and many drugs.